

Stage 1 collection

Randomised scene per episode

task ~ 8 RoboTwin-2.0 bimanual tasks
embodiment ~ piper / Franka / ARX-X5 / UR5
obstacle ~ woodenblock / pencup
spawned on the pick->place corridor, $t=0.6$

Obstacle-blind expert rollout

cuRobo MPC: plan, run $k=60$, replan
planner world holds the table only, so the expert drives straight through the obstacle
5% random $d\theta$ kick at window starts
no termination on violation

Replay buffer (memmap)

per 20 Hz tick: frozen tokens, body features, measured $d\theta$, privileged h
zero-padded to 16 state tokens, 18 actions,
1200 scene tokens; ~630 KB per step

Stage 2 HJ-SAC

Off-buffer HJ-SAC

trained

batch 64, Adam $3e-4$, gradient clip 10
204 800 steps = 200 epochs x 1024
three optimisers: critics / encoder / actor
the encoder is shaped by L_Q only; the actor sees detached features

What is learned and what is not

- ☐ HoloBrain-0 vision tower + robot-state encoder
- ☐ FK, Jacobians, $d\theta_{\max}$, and h itself
- ☒ injection, scene adapter, trunk, actor, twin critic
- ☐ V is the only learned object; h is always computed

Stage 3 deployment

Least-restrictive switch

$V(s) \geq \text{margin_on} \rightarrow a = a_{\text{nominal}}$
 $V(s) < \text{margin_on} \rightarrow a = \pi(s), \text{det.}$
release above 0.005 m (hysteresis)

Applied per replanning window

decision at each window boundary
safe actor drives the whole 0.25 s
nominal replans from the steered state

Evaluation metrics

violation rate and min h
intervention rate, task success
fraction of steps with $V \leq h$